

- 4 (a) Fig. 4.1 is a scanning electron micrograph of a section of a plant cell wall.

In living plant tissue cytoplasmic strands form part of structure **W**.

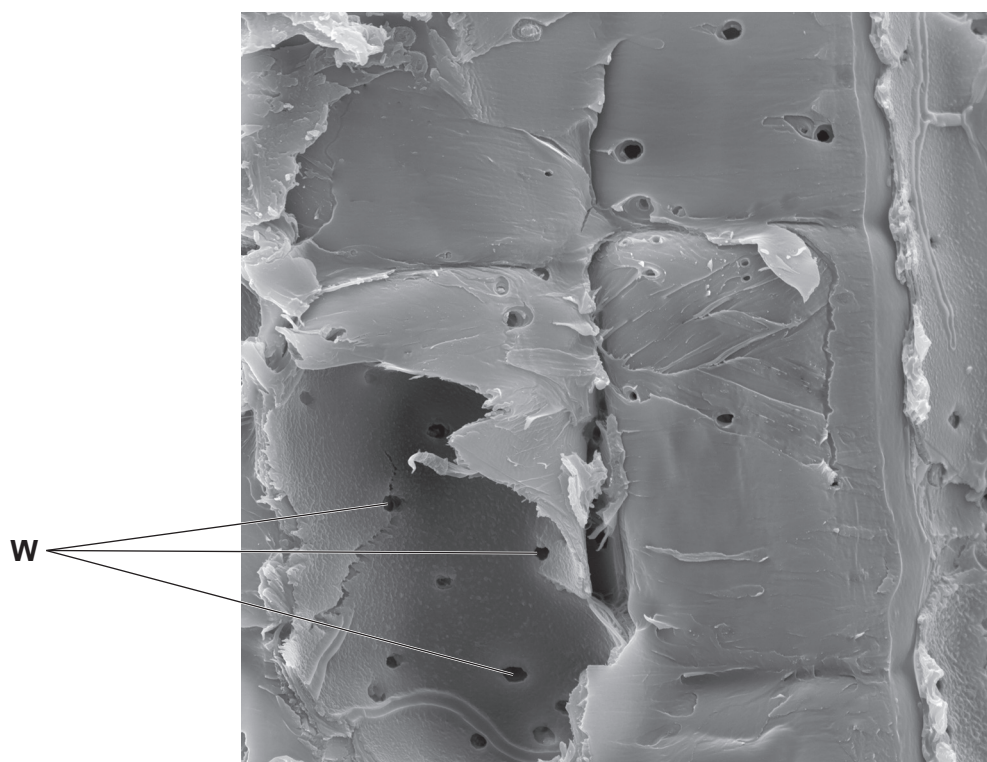


Fig. 4.1

- (i) Identify the structures labelled **W** in Fig. 4.1.

..... [1]

- (ii) Describe the function of structure **W**.

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..... [2]

- (b) Viruses can infect plant cells.

- (i) Outline the key structural features of a virus.

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..... [2]

- (ii) Many plant viruses can travel through the plant to enter companion cells in the phloem tissue. The viruses then travel with assimilates in the phloem sap to other areas of the plant.

Explain the mechanism that allows assimilates and viruses to travel through phloem sieve tubes to other areas of the plant.

..... [5]

[Total: 10]